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CBSE CLASS 10 | MATHEMATICS

COORDINATE GEOMETRY

Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Plot the geometry. Project the displacements.
Calculate with absolute certainty.

Cartesian Plane | Distance Formula | Section Formula | Polygons | Board Traps

A student-first Maths Manthan conceptual and visual revision resource



What This Handbook Includes

- Cartesian coordinate plane anatomy and projections
- Pythagorean derivation of the Distance Formula
- Geometric classification of triangles and quadrilaterals
- Rigorous similar-triangles derivation of Section Formula
- The $k:1$ parameterization and axis-intersection models
- Trisection mechanics, parallelogram diagonals, and centroid

Pedagogical Strategy: Foundations First

Coordinate geometry bridges algebra and Euclidean geometry. Every coordinate (x, y) represents signed perpendicular displacements from perpendicular reference axes. Never blindly memorize:

$$\text{Distance} = \text{Hypotenuse of Right } \triangle$$

$$\text{Section Point} = \text{Weighted Center}$$

Always draw a rough sketch with signs before setting up equations!

CBSE 2026–27 Curriculum Verification & Rationalized Scope

Official Scope (Prescribed & Tested): Concepts of coordinate geometry, graphs of linear equations, distance formula, section formula (internal division only), midpoint, and coordinates of points dividing a line segment in a given ratio. Geometrical applications to triangles and quadrilaterals.

Excluded / Deleted Topics (Non-Examinable):

- **Area of a Triangle:** $\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|$ is **deleted from the syllabus**.
- **Collinearity via Area = 0:** Must NOT be used. Collinearity must be proved using the **Distance Sum Property** ($AB+BC=AC$) or the **Consistent Section Ratio Method**.
- **External Division:** Only internal division ($m_1, m_2 > 0$) is examinable.

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Prepared for learning and conceptual mastery – Strictly aligned with CBSE Class 10 Syllabus 2026–27.

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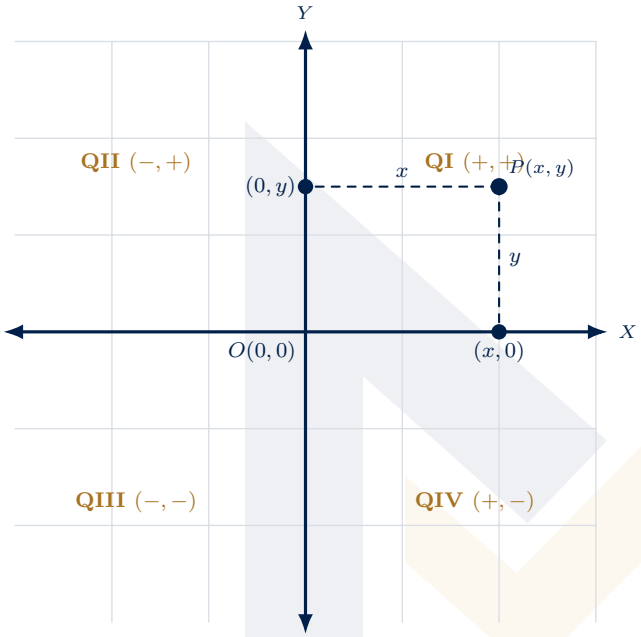
CORE SYLLABUS

COORDINATE GEOMETRY

Module 1 | Foundations: Cartesian Plane Anatomy & Distance from Axes

1.1 The Two-Dimensional Cartesian Coordinate Plane

The Cartesian coordinate system specifies each point uniquely in a plane by a pair of numerical coordinates, which are the signed distances to the point from two fixed perpendicular directed lines: the x-**axis** (horizontal) and the y-**axis** (vertical).



Fundamental Anatomy of an Ordered Pair (x, y)**:**

- **Abscissa (x-coordinate):** The signed horizontal distance of point P from the vertical y-axis.
- **Ordinate (y-coordinate):** The signed vertical distance of point P from the horizontal x-axis.
- **Points on Coordinate Axes:**
  - Any point on the x-**axis** has ordinate zero: $(\mathbf{x}, \mathbf{0})$.
  - Any point on the y-**axis** has abscissa zero: $(\mathbf{0}, \mathbf{y})$.
- **Distance of $P(x, y)$ from Coordinate Axes:**

  Distance of $\mathbf{P(x, y)}$ from **x**-axis $= |y| = |\text{ordinate}|$

  Distance of $\mathbf{P(x, y)}$ from **y**-axis $= |x| = |\text{abscissa}|$

1.2 The Absolute Value Distance Axiom

TRAP ALERT

Distance is Always Non-Negative! Coordinates can be negative, but geometric distance is **strictly non-negative**:

- The distance of the point $P(-4, -7)$ from the x-axis is $|-7| = 7$ **units**, NOT -7.
- The distance of the point $P(-4, -7)$ from the y-axis is $|-4| = 4$ **units**, NOT -4.
- In CBSE MCQs, examiners frequently include negative options to trap students who forget absolute values.

Worked Archetype 1 | Axis Projections and Distance Identification

Problem: (i) Write the coordinates of the foot of the perpendicular drawn from $P(-5, 8)$ to the y-axis. (ii) What is the distance of P from the x-axis? (iii) In which quadrant does $Q(a, b)$ lie if $a < 0$ and $b > 0$?

Solution:

1. The foot of the perpendicular from $P(-5, 8)$ to the y-axis lies directly on the y-axis, sharing the same y-coordinate. Hence, its coordinates are $(\mathbf{0}, \mathbf{8})$.
2. Distance of $P(-5, 8)$ from the x-axis $= |y| = |8| = \mathbf{8}$ units.
3. For $Q(a, b)$, since abscissa $a < 0$ (negative) and ordinate $b > 0$ (positive), point Q lies in **Quadrant II (QII)**.

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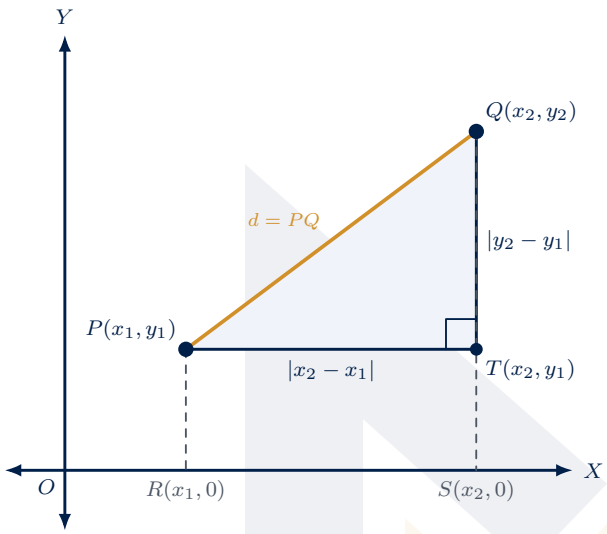
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COORDINATE GEOMETRY

Module 2 | The Distance Formula: Pythagorean Derivation & Origin Projections

2.1 Geometric Proof via the Pythagorean Theorem

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be any two points in the Cartesian plane. Draw PR and QS perpendicular to the x-axis, and draw PT perpendicular to QS.



Step-by-Step Derivation:

1. Horizontal projection: $PT = RS = OS - OR = x_2 - x_1$.
2. Vertical projection: $QT = QS - TS = QS - PR = y_2 - y_1$.
3. In right-angled triangle $\triangle PTQ$, by Pythagoras Theorem:
$$PQ^2 = PT^2 + QT^2$$
$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$
4. Taking the positive square root (since distance ≥ 0):
$$\mathbf{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

2.2 Symmetry Properties & Distance from the Origin

Essential Operational Invariants:

- **Symmetry of Direction:** Since $(a-b)^2 = (b-a)^2$, the order of points is completely irrelevant:
$$\mathbf{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}}$$
- **Distance of Any Point $P(x, y)$ from the Origin $O(0, 0)$:**
$$\mathbf{OP = \sqrt{(x - 0)^2 + (y - 0)^2} = \sqrt{x^2 + y^2}}$$

Worked Archetype 2 | Evaluating Distance with Radical and Negative Coordinates

Problem: Find the distance between: (i) $P(-6, 7)$ and $Q(-1, -5)$, and (ii) $A(a\cos\theta, 0)$ and $B(0, a\sin\theta)$.
Solution (i): Here $(x_1, y_1) = (-6, 7)$ and $(x_2, y_2) = (-1, -5)$.

$$PQ = \sqrt{[-1-(-6)]^2 + [-5-7]^2} = \sqrt{(-1+6)^2 + (-12)^2} = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = \mathbf{13} \text{ units.}$$

Solution (ii):

$$AB = \sqrt{(0 - a\cos\theta)^2 + (a\sin\theta - 0)^2} = \sqrt{a^2\cos^2\theta + a^2\sin^2\theta} = \sqrt{a^2(\cos^2\theta + \sin^2\theta)} = \sqrt{a^2(1)} = |\mathbf{a}| \text{ units.}$$

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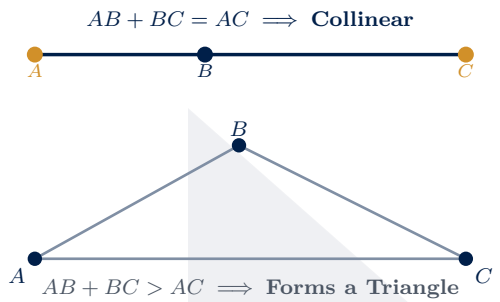
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COORDINATE GEOMETRY

Module 3 | Collinearity & Geometric Triangle Classification

3.1 Testing Collinearity Without the Area Formula

Under the rationalized CBSE 2026–27 curriculum, the area of a triangle formula is deleted. Collinearity of three points A, B, C must be established using the **Distance Sum Property**:



The Collinearity Criterion: Three distinct points A, B, and C are **collinear** (lie on the same straight line) if and only if:

Sum of two smaller distances = The largest distance

That is, one of the following holds:

- $AB + BC = AC \iff B$ lies between A and C.
- $AB + AC = BC \iff A$ lies between B and C.
- $BC + AC = AB \iff C$ lies between A and B.

If the sum of any two is strictly greater than the third, they form a non-degenerate triangle.

3.2 The 4-Way Triangle Classifier (Sides & Angles)

To determine the nature of a triangle formed by three given vertices A, B, C:

1. Compute the three side lengths: $a = BC$, $b = CA$, and $c = AB$ using the distance formula.
2. Apply the classification criteria below:

Triangle Type	Geometric Condition	Computational Check via Distance Formula
Equilateral Triangle	All 3 sides are equal	$AB = BC = CA$
Isosceles Triangle	Any 2 sides are equal	$AB = BC \neq CA$ (or any two equal)
Scalene Triangle	All 3 sides are unequal	$AB \neq BC \neq CA$
Right-Angled Triangle	Satisfies Pythagoras Converse	$(\text{Smallest side})^2 + (\text{Middle side})^2 = (\text{Largest side})^2$
Isosceles Right $\triangle$	2 sides equal + Pythagoras	$a^2 + a^2 = c^2 \iff 2a^2 = c^2$ where c is hypotenuse

Worked Archetype 3 | Proving an Isosceles Right-Angled Triangle

Problem: Show that the points $A(3, 0)$, $B(6, 4)$, and $C(-1, 3)$ are the vertices of a right-angled isosceles triangle.

Step 1: Compute the squares of the three side lengths to avoid premature square roots:

$$AB^2 = (6-3)^2 + (4-0)^2 = 3^2 + 4^2 = 9 + 16 = \mathbf{25} \implies AB = 5$$

$$BC^2 = (-1-6)^2 + (3-4)^2 = (-7)^2 + (-1)^2 = 49 + 1 = \mathbf{50} \implies BC = \sqrt{50} = 5\sqrt{2}$$

$$CA^2 = (3-(-1))^2 + (0-3)^2 = 4^2 + (-3)^2 = 16 + 9 = \mathbf{25} \implies CA = 5$$

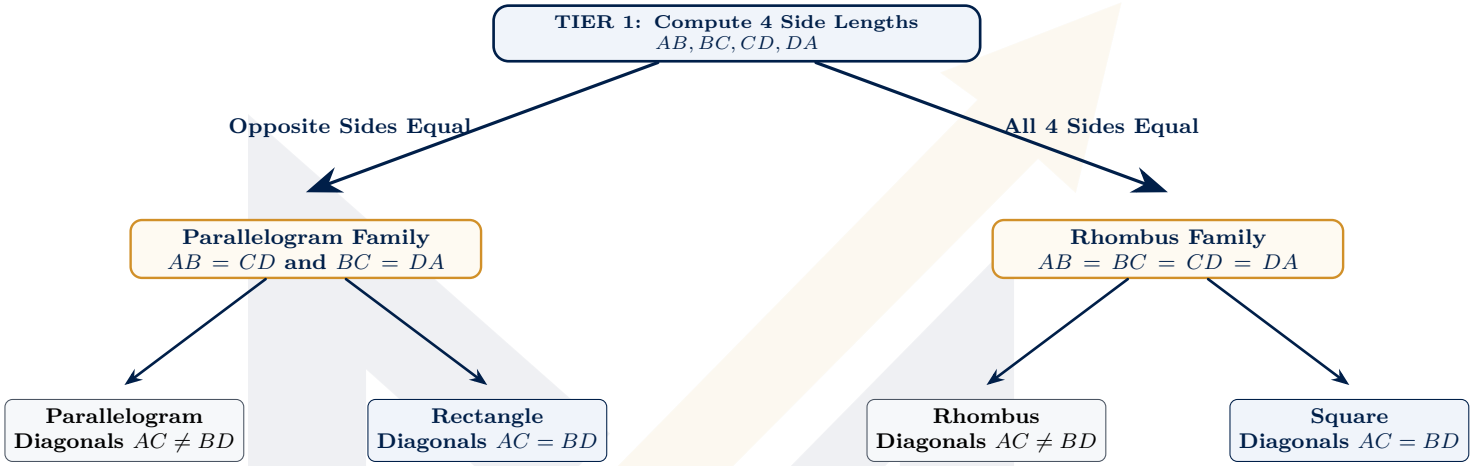
Step 2: Check side equality: $AB = CA = 5 \implies \triangle ABC$ is **isosceles**.
Step 3: Check Pythagoras Converse: $AB^2 + CA^2 = 25 + 25 = 50 = BC^2 \implies \angle A = 90°$.
Hence, $\triangle ABC$ is an **isosceles right-angled triangle** with hypotenuse BC.

COORDINATE GEOMETRY

Module 4 | Quadrilateral Identification: The 2-Tier Diagnostic Hierarchy

4.1 The 2-Tier Diagnostic Protocol for Four Given Vertices

To rigorously identify the exact geometrical nature of a quadrilateral $ABCD$ given four vertices in cyclic order, **evaluating side lengths alone is never sufficient**. You must execute the 2-Tier Protocol:



4.2 The Complete 6-Distance Verification Matrix

Quadrilateral	Tier 1: Side Length Conditions	Tier 2: Diagonal Conditions	CBSE Board Verdict
Parallelogram	$AB = CD$ and $BC = DA$	$AC \neq BD$	Diagonals unequal $\implies$ not a rectangle.
Rectangle	$AB = CD$ and $BC = DA$	**AC = BD**	Equal diagonals prove right angles.
Rhombus	$AB = BC = CD = DA$	$AC \neq BD$	Unequal diagonals $\implies$ not a square.
Square	$AB = BC = CD = DA$	**AC = BD**	Equal diagonals prove all angles $90°$.

Worked Archetype 4 | Proving Four Points Form a Rhombus but Not a Square

Problem: Show that $A(3, 0)$, $B(4, 5)$, $C(-1, 4)$, and $D(-2, -1)$ are the vertices of a rhombus. Is it a square?

Tier 1: Calculate the 4 side lengths:

- $AB^2 = (4-3)^2 + (5-0)^2 = 1^2 + 5^2 = 26 \implies AB = \sqrt{26}$
- $BC^2 = (-1-4)^2 + (4-5)^2 = (-5)^2 + (-1)^2 = 26 \implies BC = \sqrt{26}$
- $CD^2 = [-2-(-1)]^2 + (-1-4)^2 = (-1)^2 + (-5)^2 = 26 \implies CD = \sqrt{26}$
- $DA^2 = [3-(-2)]^2 + [0-(-1)]^2 = 5^2 + 1^2 = 26 \implies DA = \sqrt{26}$

Since $AB = BC = CD = DA = \sqrt{26}$, the quadrilateral $ABCD$ is a **rhombus**.

Tier 2: Calculate the 2 diagonal lengths:

- $AC^2 = (-1-3)^2 + (4-0)^2 = (-4)^2 + 4^2 = 16 + 16 = 32 \implies AC = \sqrt{32} = 4\sqrt{2}$
- $BD^2 = (-2-4)^2 + (-1-5)^2 = (-6)^2 + (-6)^2 = 36 + 36 = 72 \implies BD = \sqrt{72} = 6\sqrt{2}$

Since diagonals **AC ≠ BD**, the internal angles are not right angles. Hence, $ABCD$ is **a rhombus, but NOT a square**.

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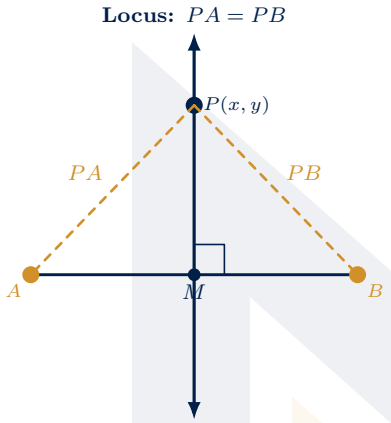
COORDINATE GEOMETRY

Module 5 | Equidistant Points, Perpendicular Bisectors & Locus

5.1 The Equidistance Principle: $PA = PB \iff PA^2 = PB^2$

A point P is equidistant from two fixed points A and B if the straight-line distances PA and PB are identical.

- **Computational Golden Rule:** Always equate the squares of the distances: $\mathbf{PA^2 = PB^2}$. This eliminates square roots immediately, preventing arithmetic errors.



Equidistant Point Configurations on Coordinate Axes:

- **Point on the x-axis:**

$$\text{Assume } \mathbf{P(x, 0)} \implies PA^2 = PB^2$$

  Solves for a single unknown x.
- **Point on the y-axis:**

$$\text{Assume } \mathbf{P(0, y)} \implies PA^2 = PB^2$$

  Solves for a single unknown y.
- **Geometric Locus (Perpendicular Bisector):** The collection of all points $P(x, y)$ equidistant from A and B forms a straight line. Expanding $PA^2 = PB^2$ always causes the quadratic terms (x^2, y^2) to cancel out, leaving a **linear equation** $Ax + By + C = 0$.

Worked Archetype 5 | Finding a Point on the x-axis Equidistant from Two Points

Problem: Find the point on the x-axis which is equidistant from $A(2, -5)$ and $B(-2, 9)$.

Step 1: Define point on x-axis: Since the point lies on the x-axis, its ordinate is zero. Let $P(x, 0)$.

Step 2: Apply the equidistance condition: $PA = PB \iff PA^2 = PB^2$.

$$(x - 2)^2 + [0 - (-5)]^2 = [x - (-2)]^2 + (0 - 9)^2$$

$$(x - 2)^2 + 5^2 = (x + 2)^2 + (-9)^2$$

Step 3: Expand and cancel quadratic terms:

$$x^2 - 4x + 4 + 25 = x^2 + 4x + 4 + 81$$

$$-4x + 29 = 4x + 85 \implies -8x = 85 - 29 = 56 \implies x = \frac{56}{-8} = \mathbf{-7}$$

Conclusion: The required point on the x-axis is $\mathbf{P(-7, 0)}$.

Worked Archetype 6 | Linear Relation between x and y (Perpendicular Bisector)

Problem: Find a relation between x and y such that the point $P(x, y)$ is equidistant from $A(3, 6)$ and $B(-3, 4)$.

$$PA^2 = PB^2 \implies (x - 3)^2 + (y - 6)^2 = [x - (-3)]^2 + (y - 4)^2$$

$$(x^2 - 6x + 9) + (y^2 - 12y + 36) = (x^2 + 6x + 9) + (y^2 - 8y + 16)$$

$$-6x - 12y + 45 = 6x - 8y + 25$$

$$-12x - 4y + 20 = 0 \implies \text{Divide by } -4: \quad \mathbf{3x + y - 5 = 0}$$

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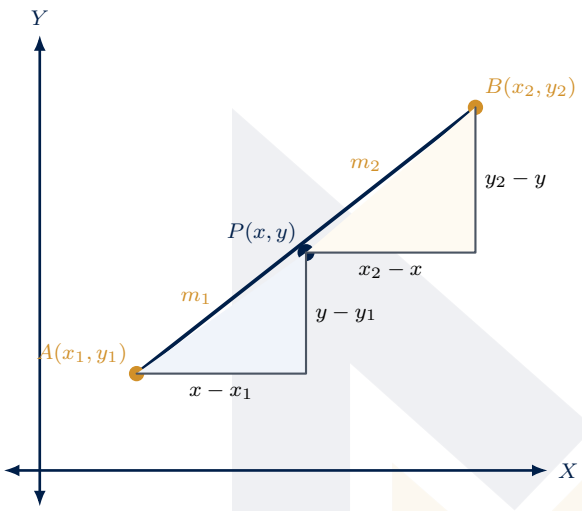
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COORDINATE GEOMETRY

Module 6 | The Section Formula: Similar Triangles Proof & Midpoint Axiom

6.1 Geometric Derivation of the Section Formula

Let $P(x, y)$ divide the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m_1 : m_2$. Draw vertical lines from A, P, B to the x-axis, and draw horizontal segments AQ and PC.



Similar Triangles Step-by-Step Proof: In $\triangle AQP$ and $\triangle PCB$, since the corresponding angles are equal ($\angle A = \angle P$, right angles at Q, C):

$$\triangle AQP \sim \triangle PCB \quad \text{(AA similarity)}$$

$$\frac{AQ}{PC} = \frac{QP}{CB} = \frac{AP}{PB} = \frac{m_1}{m_2}$$

Equating horizontal ratio:

$$\frac{x - x_1}{x_2 - x} = \frac{m_1}{m_2} \implies (m_1 + m_2)x = m_1x_2 + m_2x_1$$

$$\mathbf{x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}}$$

Similarly, equating vertical ratios yields:

$$\mathbf{y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}}$$

6.2 The Master Section Formula Suite & The Midpoint Special Case

The Section Formula for Internal Division (P divides AB in ratio $m_1 : m_2$):

$$\mathbf{P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)}$$

The Midpoint Formula ($1 : 1$ Division, where $m_1 = m_2 = 1$):

$$\mathbf{M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)}$$

Worked Archetype 7 | Direct Application of Section and Midpoint Formulas

Problem: (i) Find coordinates of point P dividing $A(4, -3)$ and $B(8, 5)$ in ratio $3 : 1$. (ii) Find midpoint M of AB.

Solution (i): Here $(x_1, y_1) = (4, -3)$, $(x_2, y_2) = (8, 5)$, $m_1 = 3$, $m_2 = 1$.

$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2} = \frac{3(8) + 1(4)}{3 + 1} = \frac{24 + 4}{4} = \frac{28}{4} = \mathbf{7}$$

$$y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2} = \frac{3(5) + 1(-3)}{3 + 1} = \frac{15 - 3}{4} = \frac{12}{4} = \mathbf{3} \implies \mathbf{P(7, 3)}$$

Solution (ii): $M = \left(\frac{4+8}{2}, \frac{-3+5}{2}\right) = \left(\frac{12}{2}, \frac{2}{2}\right) = \mathbf{(6, 1)}$.

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Module 7 | The $k:1$ Ratio Parameterization & Coordinate Axis Division

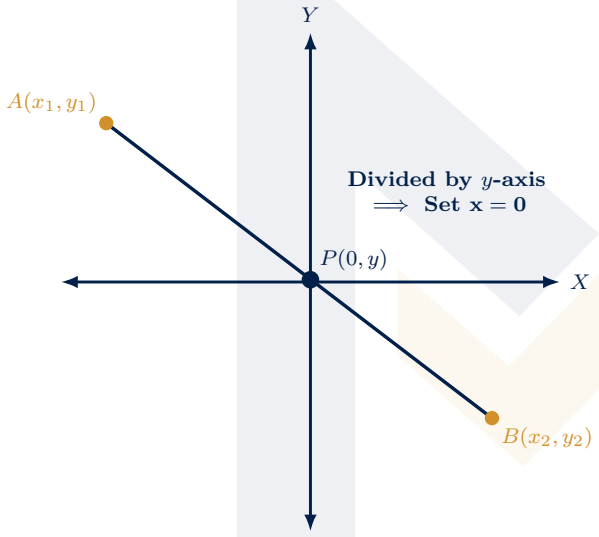
7.1 Why the $k:1$ Method is Superior to $m_1:m_2$

When the dividing ratio is unknown, writing $m_1:m_2$ introduces **two variables**, requiring system equations. Setting $\mathbf{k=\frac{m_1}{m_2}}$ reduces the problem to a **single unknown** k:

$$\mathbf{P(x,y)=\left(\frac{kx_2+x_1}{k+1},\ \frac{ky_2+y_1}{k+1}\right)}$$

Once k is found, the required ratio is $\mathbf{k:1}$ (e.g., if $k=\frac{2}{3}$, ratio is $\frac{2}{3}:1 \implies \mathbf{2:3}$).

7.2 Division of a Line Segment by Coordinate Axes



The 2-Case Axis Intersection Rules:

- **Case 1: Line Divided by the x-axis:**
  1. Intersection point lies on x-axis: $\mathbf{P(x,0)}$.
  2. Its y-coordinate is zero: set $\frac{\mathbf{ky_2+y_1}}{\mathbf{k+1}}=\mathbf{0}$.
  3. Yields: $ky_2+y_1=0 \implies \mathbf{k=-\frac{y_1}{y_2}}$.
- **Case 2: Line Divided by the y-axis:**
  1. Intersection point lies on y-axis: $\mathbf{P(0,y)}$.
  2. Its x-coordinate is zero: set $\frac{\mathbf{kx_2+x_1}}{\mathbf{k+1}}=\mathbf{0}$.
  3. Yields: $kx_2+x_1=0 \implies \mathbf{k=-\frac{x_1}{x_2}}$.

Worked Archetype 8 | Division of Line Segment by the y-axis

Problem: Find the ratio in which the y-axis divides the line segment joining the points $A(5,-6)$ and $B(-1,-4)$. Also find the point of intersection.

Step 1: Set up the $k:1$ parameterization: Let the y-axis divide AB in the ratio $k:1$ at point $P(0,y)$.

$$P(x,y)=\left(\frac{k(-1)+1(5)}{k+1},\ \frac{k(-4)+1(-6)}{k+1}\right)=\left(\frac{-k+5}{k+1},\ \frac{-4k-6}{k+1}\right)$$

Step 2: Equate abscissa to zero: Since P lies on the y-axis, $x=0$:

$$\frac{-k+5}{k+1}=0 \implies -k+5=0 \implies \mathbf{k=5}$$

Thus, the required dividing ratio is $5:1 \implies \mathbf{5:1}$ internally.

Step 3: Find the y-coordinate of intersection:

$$y=\frac{-4(5)-6}{5+1}=\frac{-20-6}{6}=\frac{-26}{6}=-\frac{\mathbf{13}}{\mathbf{3}}$$

Conclusion: The y-axis divides AB in the ratio $\mathbf{5:1}$, and the point of intersection is $\mathbf{\left(0,-\frac{13}{3}\right)}$.

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Module 8 | Points of Trisection & Multi-Segment Partitioning

8.1 The Geometry of Trisection Points

The **points of trisection** of a line segment AB are the two distinct points, say P and Q, which divide the segment into three equal parts, such that $\mathbf{AP = PQ = QB}$.



Master Ratios for Trisection Points:

- **Point P:** Divides AB internally in the ratio $\mathbf{1 : 2}$.

$$\mathbf{P} = \left(\frac{\mathbf{1(x_2) + 2(x_1)}}{\mathbf{1+2}}, \frac{\mathbf{1(y_2) + 2(y_1)}}{\mathbf{1+2}}\right) = \left(\frac{\mathbf{x_2 + 2x_1}}{\mathbf{3}}, \frac{\mathbf{y_2 + 2y_1}}{\mathbf{3}}\right)$$

- **Point Q:** Divides AB internally in the ratio $\mathbf{2 : 1}$.

$$\mathbf{Q} = \left(\frac{\mathbf{2(x_2) + 1(x_1)}}{\mathbf{2+1}}, \frac{\mathbf{2(y_2) + 1(y_1)}}{\mathbf{2+1}}\right) = \left(\frac{\mathbf{2x_2 + x_1}}{\mathbf{3}}, \frac{\mathbf{2y_2 + y_1}}{\mathbf{3}}\right)$$

- **The High-Speed Midpoint Optimization:** Do NOT apply the full section formula twice! Once point P is determined, observe that Q is the **midpoint of segment** PB:

$$\mathbf{Q} = \text{Midpoint of } \mathbf{PB} = \left(\frac{\mathbf{x_P + x_2}}{\mathbf{2}}, \frac{\mathbf{y_P + y_2}}{\mathbf{2}}\right)$$

8.2 Partitioning into Four Equal Parts (Quadrisection)

To divide a segment AB into 4 equal segments:

1. First, find the overall midpoint of AB, say $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
2. Next, find the midpoint of AM, say P_1, and the midpoint of MB, say P_2.

Worked Archetype 9 | Finding the Coordinates of the Points of Trisection

Problem: Find the coordinates of the points of trisection of the line segment joining $A(2, -2)$ and $B(-7, 4)$.

Step 1: Point P divides AB in ratio $1 : 2$ ($m_1 = 1, m_2 = 2$):

$$x_P = \frac{1(-7) + 2(2)}{1+2} = \frac{-7+4}{3} = \frac{-3}{3} = \mathbf{-1}$$

$$y_P = \frac{1(4) + 2(-2)}{1+2} = \frac{4-4}{3} = \frac{0}{3} = \mathbf{0} \implies \mathbf{P(-1, 0)}$$

Step 2: Use the Midpoint Optimization to find Q (Midpoint of PB): Here $P = (-1, 0)$ and $B = (-7, 4)$:

$$x_Q = \frac{-1 + (-7)}{2} = \frac{-8}{2} = \mathbf{-4}$$

$$y_Q = \frac{0+4}{2} = \frac{4}{2} = \mathbf{2} \implies \mathbf{Q(-4, 2)}$$

Conclusion: The points of trisection are $\mathbf{P(-1, 0)}$ and $\mathbf{Q(-4, 2)}$.

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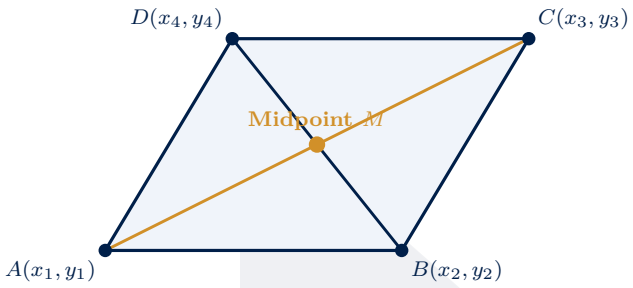
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Module 9 | The Parallelogram Diagonal Trick & Centroid of a Triangle

9.1 The 10-Second Parallelogram Fourth-Vertex Technique

In Euclidean geometry, **the diagonals of any parallelogram bisect each other**. Therefore, the diagonals AC and BD have the exact same midpoint M:



Algebraic Parallelogram Identity:

$$\text{Midpoint of } AC = \text{Midpoint of } BD$$

$$\left(\frac{x_1+x_3}{2}, \frac{y_1+y_3}{2}\right) = \left(\frac{x_2+x_4}{2}, \frac{y_2+y_4}{2}\right)$$

Equating coordinates yields the instantaneous formula:

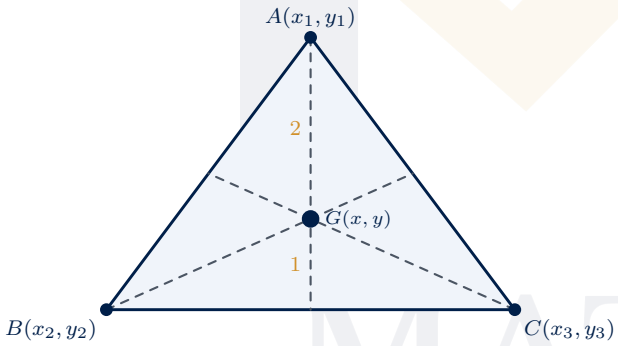
$$\mathbf{x_1 + x_3 = x_2 + x_4 \iff x_4 = x_1 + x_3 - x_2}$$

$$\mathbf{y_1 + y_3 = y_2 + y_4 \iff y_4 = y_1 + y_3 - y_2}$$

Avoids cumbersome distance formulas entirely!

9.2 Centroid of a Triangle (Center of Mass)

The **centroid** (G) of a triangle is the point of intersection of its three medians. It divides every median internally in the ratio $\mathbf{2:1}$ (measured from vertex to base midpoint):



Centroid Coordinates Formula: Since D is the midpoint of $BC \implies D = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$. G divides AD in ratio $2:1$. Applying section formula:

$$x = \frac{2\left(\frac{x_2+x_3}{2}\right) + 1(x_1)}{2+1} = \frac{x_1+x_2+x_3}{3}$$

$$\mathbf{G(x,y) = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)}$$

Worked Archetype 10 | Finding Unknown Vertex of a Parallelogram

Problem: If $(1,2)$, $(4,y)$, $(x,6)$, and $(3,5)$ are the vertices of a parallelogram taken in cyclic order, find x and y.
Solution: Let $A(1,2)$, $B(4,y)$, $C(x,6)$, and $D(3,5)$. Diagonals AC and BD bisect each other:

$$\text{Midpoint of } AC = \left(\frac{1+x}{2}, \frac{2+6}{2}\right) = \left(\frac{1+x}{2}, 4\right)$$

$$\text{Midpoint of } BD = \left(\frac{4+3}{2}, \frac{y+5}{2}\right) = \left(\frac{7}{2}, \frac{y+5}{2}\right)$$

Equating x-coordinates: $\frac{1+x}{2} = \frac{7}{2} \implies 1+x=7 \implies \mathbf{x=6}$.
Equating y-coordinates: $\frac{y+5}{2} = 4 \implies y+5=8 \implies \mathbf{y=3}$.

CHAPTER 07

QUICK REFERENCE

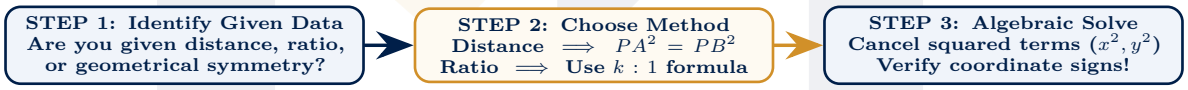
COORDINATE GEOMETRY

Module 10 | Master Formula Matrix & Decision Protocol

11.1 The Comprehensive Mensuration Matrix for Coordinate Geometry

Formula Concept	Inputs	Mathematical Expression	Strategic Board Purpose
Distance Formula	$(x_1, y_1), (x_2, y_2)$	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	Distance between 2 points; side lengths.
Distance from Origin	(x, y)	$d = \sqrt{x^2 + y^2}$	Special case when one point is $(0, 0)$.
Distance from x-axis	(x, y)	Distance $= \lvert y \rvert = \lvert$ordinate$\rvert$	Absolute vertical distance (MCQ favorite).
Distance from y-axis	(x, y)	Distance $= \lvert x \rvert = \lvert$abscissa$\rvert$	Absolute horizontal distance.
Section Formula	$m_1 : m_2$	$\left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}\right)$	Internal division coordinates.
$k : 1$ Ratio Form	$k : 1$	$\left(\frac{kx_2+x_1}{k+1}, \frac{ky_2+y_1}{k+1}\right)$	Finding unknown dividing ratios.
Midpoint Formula	$1 : 1$	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$	Center of circle; trisection shortcut.
Parallelogram Vertex	A, B, C, D	$x_1 + x_3 = x_2 + x_4,\ y_1 + y_3 = y_2 + y_4$	10-second fourth vertex calculation.
Triangle Centroid	3 Vertices	$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$	Center of mass / medians concurrency.

11.2 The 3-Step Board Examination Decision Flowchart



Fast Mental Calculation Checks for Board Exams:

- **Pythagorean Triplet Recognition:** Differences $|x_2 - x_1|$ and $|y_2 - y_1|$ matching $(3, 4) \implies d = 5$; $(5, 12) \implies d = 13$; $(7, 24) \implies d = 25$; $(8, 15) \implies d = 17$.
- **Circle Diameter Center:** The center of a circle is always the **midpoint of its diameter**: $C = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. If center and one endpoint are given: $x_2 = 2x_C - x_1$, $y_2 = 2y_C - y_1$.

MATHS MANTHAN TUITION CLASSES

COORDINATE GEOMETRY

Module 11 | Board Exam Archetypes: 3-Mark & 4-Mark Stepwise Layout

Archetype A (3-Mark Board Model) | Circle Diameter & Missing Endpoint

Archetype A: Center of Circle as Diameter Midpoint

Problem: Find the coordinates of a point A, where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.

Step 1: State the geometric property [1 Mark]:
The centre of a circle C is the midpoint of its diameter AB. Let the coordinates of point A be (x, y). The given endpoint is $B(1, 4)$, and the centre is $C(2, -3)$.

Step 2: Apply the midpoint formula [1 Mark]:

$$C(x_C, y_C) = \left(\frac{x+1}{2}, \frac{y+4}{2}\right) = (2, -3)$$

Step 3: Solve for x and y independently [1 Mark]:

$$\frac{x+1}{2} = 2 \implies x + 1 = 4 \implies \mathbf{x = 3}$$

$$\frac{y+4}{2} = -3 \implies y + 4 = -6 \implies \mathbf{y = -10}$$

Final Statement: The coordinates of point A are $(\mathbf{3}, \mathbf{-10})$.

Archetype B (4-Mark Board Model) | Section Formula with $AP = \frac{3}{7}AB$

Archetype B: Ratio Decoding and Coordinate Determination

Problem: If $A(-2, -2)$ and $B(2, -4)$ are two points, find the coordinates of P such that $AP = \frac{3}{7}AB$ and P lies on the line segment AB.

Step 1: Decode the internal division ratio $m_1 : m_2$ [1.5 Marks]:
We are given that $AP = \frac{3}{7}AB$. Since P lies on the segment AB:

$$PB = AB - AP = AB - \frac{3}{7}AB = \frac{4}{7}AB$$

Now compute the ratio of the parts:

$$\frac{AP}{PB} = \frac{\frac{3}{7}AB}{\frac{4}{7}AB} = \frac{3}{4} \implies \mathbf{m_1 : m_2 = 3 : 4}$$

TRAP ALERT

Common Error Alert: Many students mistakenly use $3 : 7$. The ratio is strictly $3 : 4$!

Step 2: Apply the section formula [1.5 Marks]: Here $(x_1, y_1) = (-2, -2)$, $(x_2, y_2) = (2, -4)$, $m_1 = 3$, $m_2 = 4$.

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} = \frac{3(2) + 4(-2)}{3 + 4} = \frac{6 - 8}{7} = -\frac{\mathbf{2}}{\mathbf{7}}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} = \frac{3(-4) + 4(-2)}{3 + 4} = \frac{-12 - 8}{7} = -\frac{\mathbf{20}}{\mathbf{7}}$$

Step 3: Concluding result [1 Mark]:
The coordinates of point P are $\left(-\frac{\mathbf{2}}{\mathbf{7}}, -\frac{\mathbf{20}}{\mathbf{7}}\right)$.

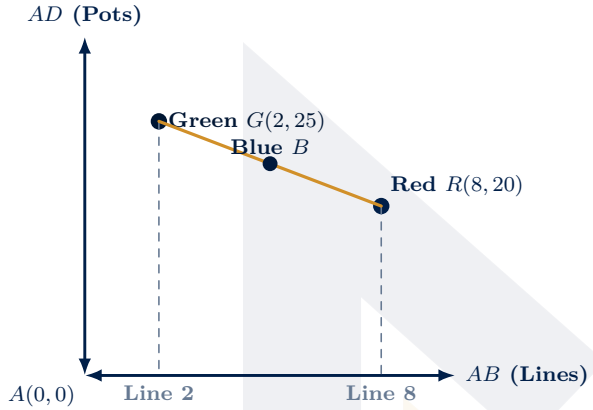
COORDINATE GEOMETRY

Module 12 | Applied Case-Study: Sports Ground Grid & GPS Navigation

Case Study 1 | Sports Day Relay Race on a Coordinate Grid

To conduct sports day activities in a rectangular school ground $ABCD$, lines have been drawn with chalk powder at a distance of 1 m each. 100 flower pots have been placed at a distance of 1 m from each other along AD, so that $AD = 100$ m.

- Niharika runs $\frac{1}{4}$th of the distance AD on the 2nd line and posts a green flag.
- Preet runs $\frac{1}{5}$th of the distance AD on the 8th line and posts a red flag.



4-Mark Board Question Protocol

Questions:

1. Write the coordinates of the green flag G and the red flag R. [1M]
2. Find the straight-line distance between both flags. [1M]
3. If Rashmi has to post a blue flag exactly halfway between the line segment joining the two flags, on which line and at what distance should she post her flag? [2M]

Solution & Marking Scheme Evaluation

1. **Coordinates of Flags [1M]:**
   - Green flag G: $x = 2$, $y = \frac{1}{4} \times 100 = 25 \implies \mathbf{G(2, 25)}$.
   - Red flag R: $x = 8$, $y = \frac{1}{5} \times 100 = 20 \implies \mathbf{R(8, 20)}$.
2. **Distance Between Flags [1M]:**

$$d = \sqrt{(8-2)^2 + (20-25)^2} = \sqrt{6^2 + (-5)^2} = \sqrt{36 + 25} = \sqrt{\mathbf{61}} \text{ m} \approx 7.81 \text{ m}$$

3. **Position of the Blue Flag [2M]:** The blue flag is posted at the **midpoint** of GR:

$$M = \left(\frac{2+8}{2}, \frac{25+20}{2}\right) = \left(\frac{10}{2}, \frac{45}{2}\right) = (\mathbf{5}, \mathbf{22.5})$$

Answer: Rashmi should post her blue flag on the **5th line** at a distance of 22.5 **m**.

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CHAPTER 07

EXAM STRATEGY

COORDINATE GEOMETRY

Module 13 | Board Traps, Verification Hacks & 5-Point Audit

13.1 The Top 7 Coordinate Geometry Board Traps (And How to Bust Them)

Trap 1: The Negative Sign Subtraction Trap: In $(x_2 - x_1)$, when x_1 is negative, writing $x_2 - x_1$ as $x_2 - 5$ instead of $x_2 - (-5) = \mathbf{x_2 + 5}$.

TRAP ALERT

Always enclose negative coordinates in parentheses!

Trap 2: The Section Formula Cross-Index Confusion: Multiplying m_1 with x_1 instead of x_2. Remember the criss-cross rule: $\mathbf{m_1}$ multiplies with the **opposite point** (x_2, y_2).

Trap 3: Axis Coordinate Mix-up: Assuming a point on the y-axis is $(y, 0)$. A point on the **y-axis has** $x = 0$, so it is strictly $(\mathbf{0}, \mathbf{y})$. A point on the x-axis is $(\mathbf{x}, \mathbf{0})$.

Trap 4: The Trisection $1:3$ Error: Believing trisection means ratio $1:3$. Three equal parts means the internal ratios are strictly $\mathbf{1:2}$ and $\mathbf{2:1}$!

Trap 5: Squaring Incomplete Binomials: Expanding $(x-3)^2$ as $x^2 - 9$ or $x^2 + 9$ instead of $\mathbf{x^2 - 6x + 9}$.

Trap 6: Parallelogram Cyclic Order Violation: Taking vertices out of order (e.g., across diagonals). Vertices A, B, C, D must be taken in **continuous cyclic order** along the perimeter.

Trap 7: The Deleted Triangle Area Formula Trap: Using the triangle area formula $\frac{1}{2}|x(y-y)+\ldots|$ to prove collinearity. This formula is **deleted from the CBSE syllabus**; use the **Distance Sum Rule** $(AB + BC = AC)$.

13.2 High-Speed Algebraic Verification Protocol

The Slow Method (High Risk of Arithmetic Slip)	The Maths Manthan Factoring Method
Finding 4th vertex of parallelogram $ABCD$:	Use the common diagonal midpoint property:
Set up distance equations: $AB = CD$ and $BC = DA$.	Midpoint(AC) = Midpoint(BD)
Leads to complicated non-linear simultaneous radical equations requiring quadratic substitution.	$x_1 + x_3 = x_2 + x_4 \implies \mathbf{x_4 = x_1 + x_3 - x_2}$
Time taken: ≈ 5 minutes.	$y_1 + y_3 = y_2 + y_4 \implies \mathbf{y_4 = y_1 + y_3 - y_2}$
Arithmetic error rate: $> 35\%$.	**Solves in 10 seconds with 100% certainty!**

13.3 The 5-Point Board Examination Pre-Submission Checklist

Final 60-Second Answer Sheet Verification Protocol:

- ☐ **Coordinates Stated with Signs:** Did you double check that $(x_2 - x_1)$ correctly handled negative signs with double parentheses?
- ☐ **Symbolic Formula Stated:** Did you write $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ or the section formula before substituting values? (Guarantees CBSE method marks).
- ☐ **Squared Distances Equated:** For equidistant questions $(PA = PB)$, did you write $PA^2 = PB^2$ to cleanly justify eliminating square roots?
- ☐ **Ratio Stated Correctly:** Did you confirm whether the question asked for $m_1 : m_2$ (parts of segment) or coordinates of the point?
- ☐ **Geometric Units Attached:** Are final linear distances labeled with units?

MATHS MANTHAN TUITION CLASSES | END OF CHAPTER 7 HANDBOOK

Designed for Academic Excellence in CBSE Class 10 Board Examinations 2026–27.